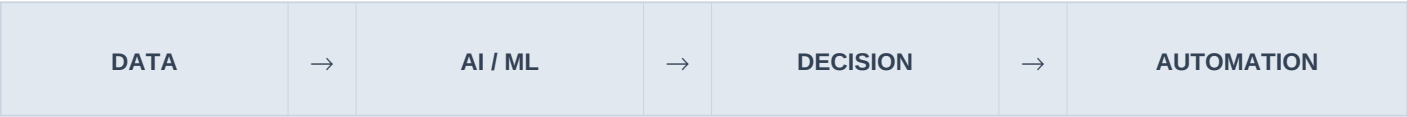


Medical Operations Intelligence Dashboard & Automation System

AWS-Centric Architecture & Project Overview



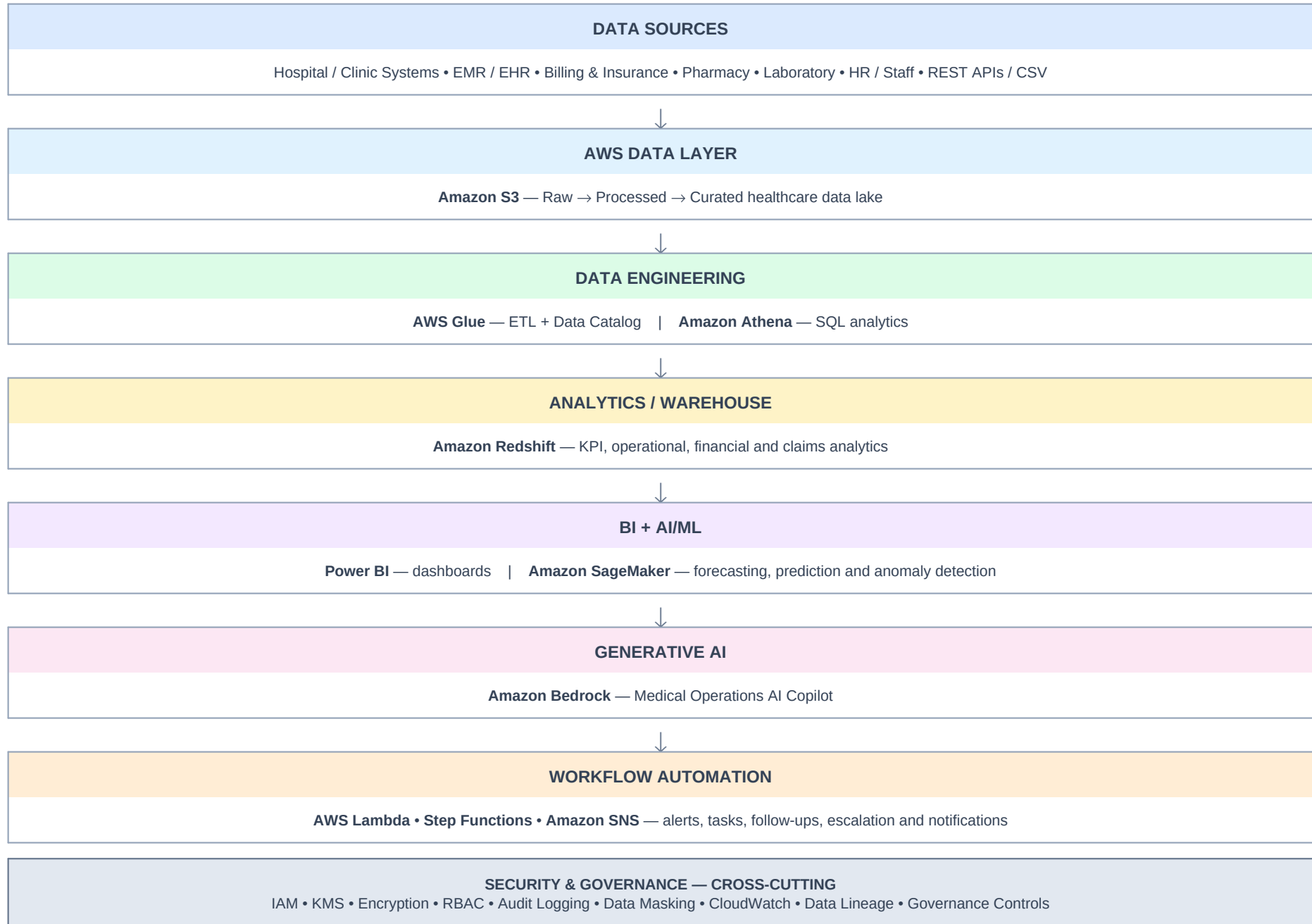
What are we building?

A cloud-based healthcare operations intelligence platform that organizes hospital/clinic data, produces operational dashboards, applies AI/ML for prediction and insight, and triggers automated actions.

How are we building it?

Incrementally: synthetic data foundation → AWS data engineering → Power BI analytics → AI/ML → AI Copilot → workflow automation.

1. High-Level AWS Architecture



2. Major Functional Modules

Module	Scope
Executive Command Center	Enterprise KPIs, facility comparison, operational health score, management scorecards and drill-down
Patient Operations	Appointments, no-shows, waiting time, patient flow, admissions, discharges and length of stay
Doctor & Staff Intelligence	Provider productivity, utilization, workload, shift insights and capacity forecasting
Billing & Claims	Revenue, collections, outstanding payments, claim status, denials, aging and payer performance
Pharmacy & Inventory	Stock, consumption, expiry, low-stock alerts, forecasting and reorder recommendations
AI & Predictive Intelligence	Forecasting, prediction, anomaly detection, root-cause analysis and recommendations
Workflow Automation	Alerts, approvals, follow-ups, task assignment, SLA monitoring and recurring workflows
AI Operations Copilot	Natural-language questions, data-backed explanations, summaries and recommended actions

3. Current Implementation Progress

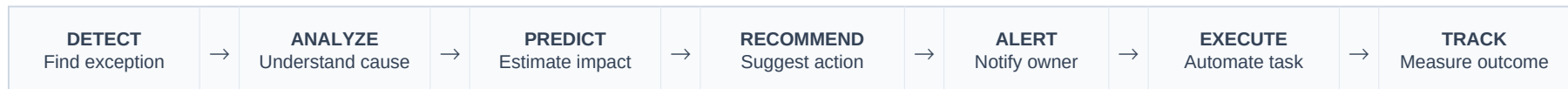
Stage	Status	Progress
Core data foundation	COMPLETED	Facilities, departments, doctors, patients and 50,000 appointments
Data validation	COMPLETED	Missing-value, duplicate, relationship and business-rule checks
Billing & revenue	COMPLETED	35,153 billing records with gross, discount, net, paid and outstanding amounts
Insurance claims	COMPLETED	26,330 claims with payer, status, denial reasons and outstanding claim amounts
Admissions & beds	NEXT	Admissions, discharges, beds and occupancy
AWS implementation	UPCOMING	S3 → Glue → Athena / warehouse → analytics
Power BI	UPCOMING	Executive and operational dashboards, DAX KPIs and drill-through
AI/ML + automation	UPCOMING	Forecasting, prediction, Bedrock Copilot and event-driven workflows

4. Core Data Model

The foundation uses relational healthcare data: master tables describe facilities, departments, doctors and patients; transaction tables capture appointments, billing and claims.



5. Intelligence & Automation Flow



Example: Low appointment utilization detected → AI identifies the affected department → demand forecast checks upcoming workload → system recommends schedule optimization → manager receives an alert → automated task is created → Power BI tracks the outcome.

6. Implementation Roadmap

Phase	Focus	Deliverables
1	Data Foundation	Synthetic healthcare datasets, relationships, validation and data dictionary
2	AWS Cloud Foundation	S3 storage, IAM, secure data organization and access
3	Data Engineering	Glue ETL, Data Catalog and Athena / warehouse analytics
4	Power BI Analytics	Executive Command Center, operational dashboards, DAX KPIs and drill-through
5	Predictive Intelligence	Patient volume, no-show, claim-denial and inventory forecasting
6	AI Copilot	Bedrock-powered operations Q&A;, summaries, explanations and recommendations
7	Automation	Lambda / Step Functions / SNS alerts, tasks, follow-ups and exception workflows
8	Final Delivery	Architecture documentation, GitHub README, presentation, demo and interview preparation

Core project principle: DATA → AI → DECISION → AUTOMATION